Differential Equations

ANSWER KEY

Section 1: Separation of Variables

1 Solve the differential equation $\frac{dy}{dx} = \frac{2x}{y}$ with initial condition $y(0) = 4$ for $y > 0$.

a. $y = \sqrt{2x^2 + 16}$ ✓

b. $y = 2x^2 + 4$

c. $y = \sqrt{x^2 + 16}$

d. $y = x^2 + 4$

Separate variables: $ydy = 2xdx$. Integrate: $\frac{1}{2}y^2 = x^2 + C \Rightarrow y^2 = 2x^2 + 2C$. Since $y(0) = 4$, $16 = 2C$. Thus $y^2 = 2x^2 + 16 \Rightarrow y = \sqrt{2x^2 + 16}$.

2 Solve $\frac{dy}{dx} = -3y$ with $y(0) = 50$.

a. $y = 50e^{-3x}$ ✓

b. $y = 50e^{3x}$

c. $y = -3x + 50$

d. $y = 50 - 3e^x$

$\frac{1}{y}dy = -3dx \Rightarrow \ln|y| = -3x + C \Rightarrow y = Ae^{-3x}$. With $y(0) = 50$, $y = 50e^{-3x}$.

3 Solve $\frac{dy}{dx} = (1 + y^2)e^x$ with $y(0) = 0$.

a. $y = \tan(e^x - 1)$ ✓

b. $y = \arctan(e^x - 1)$

c. $y = \tan(e^x)$

d. $y = e^x - 1$

$\frac{1}{1+y^2}dy = e^x dx \Rightarrow \arctan y = e^x + C$. With $y(0) = 0$, $\arctan 0 = e^0 + C \Rightarrow 0 = 1 + C \Rightarrow C = -1$. Thus $\arctan y = e^x - 1 \Rightarrow y = \tan(e^x - 1)$.

Section 2: Integrating Factor Method for First-Order Linear ODEs

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4 For the linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$, what is the integrating factor $I(x)$?

a. $I(x) = e^{\int P(x)dx}$ ✓

b. $I(x) = \int P(x)dx$

c. $I(x) = e^{-\int P(x)dx}$

d. $I(x) = e^{\int Q(x)dx}$

Multiplying by $I(x) = e^{\int P(x)dx}$ collapses the left-hand side into the product rule derivative $\frac{d}{dx}[I(x)y]$.

5 Find the integrating factor for $\frac{dy}{dx} + \frac{3}{x}y = x^2$ for $x > 0$.

a. x^3 ✓

b. $3\ln x$

c. e^{3x}

d. $\frac{1}{x^3}$

$I(x) = e^{\int (3/x)dx} = e^{3\ln x} = e^{\ln(x^3)} = x^3$.

6 Solve $\frac{dy}{dx} + 2y = 4$ with $y(0) = 5$.

a. $y = 2 + 3e^{-2x}$ ✓

b. $y = 2 + 5e^{-2x}$

c. $y = 4 + e^{-2x}$

d. $y = 2 - 3e^{-2x}$

$I(x) = e^{2x}$. Multiplying: $\frac{d}{dx}[e^{2x}y] = 4e^{2x} \Rightarrow e^{2x}y = 2e^{2x} + C \Rightarrow y = 2 + Ce^{-2x}$. Using $y(0) = 5 \Rightarrow 5 = 2 + C \Rightarrow C = 3$. Thus $y = 2 + 3e^{-2x}$.

Section 3: Homogeneous Differential Equations (y = vx)

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7 Which substitution transforms a homogeneous differential equation $\frac{dy}{dx} = f(y/x)$ into a separable equation?

a. $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ ✓

b. $y = v + x$

c. $v = xy$

d. $y = v^2 x$

Let $v = y/x \Rightarrow y = vx$. By the product rule, $\frac{dy}{dx} = v + x\frac{dv}{dx}$, reducing the ODE to $v + x\frac{dv}{dx} = f(v)$.

8 Classify the differential equation $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$.

a. **Homogeneous of degree 0** ✓

b. First-order linear

c. Exact non-linear

d. Second-order Bernoulli

Dividing numerator and denominator by x^2 gives $\frac{1 + (y/x)^2}{2(y/x)}$, which is a function purely of y/x, classifying it as homogeneous.

9 Using $y = vx$, what does $\frac{dy}{dx} = \frac{y}{x} + 1$ become?

a. $x\frac{dv}{dx} = 1$ ✓

b. $v + \frac{dv}{dx} = 1$

c. $x\frac{dv}{dx} = v + 1$

d. $\frac{dv}{dx} = x$

Substitute $\frac{dy}{dx} = v + x\frac{dv}{dx}$ and $y/x = v$: $v + x\frac{dv}{dx} = v + 1 \Rightarrow x\frac{dv}{dx} = 1$.

Section 4: Numerical Euler's Method

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10 What is Euler's recurrence relation for approximating solutions to $\frac{dy}{dx} = f(x, y)$ with step size h?

a. $y_{n+1} = y_n + hf(x_n, y_n)$ ✓

b. $y_{n+1} = y_n + \frac{h}{2}f(x_n, y_n)$

c. $y_{n+1} = f(x_n + h, y_n)$

d. $y_{n+1} = y_n + h^2 f(x_n, y_n)$

Euler's method uses the tangent line approximation: $y_{n+1} = y_n + h\frac{dy}{dx}\Big|_{(x_n, y_n)} = y_n + hf(x_n, y_n)$.

11 Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with step size $h = 0.1$ to find y_1 at $x = 0.1$.

a. 1.10 ✓

b. 1.05

c. 1.21

d. 1.15

$y_1 = y_0 + h(x_0 + y_0) = 1 + 0.1(0 + 1) = 1 + 0.1 = 1.10$.

12 Using the result above ($x_1 = 0.1, y_1 = 1.10$), find y_2 at $x = 0.2$ with $h = 0.1$.

a. 1.22 ✓

b. 1.20

c. 1.25

d. 1.31

$y_2 = y_1 + h(x_1 + y_1) = 1.10 + 0.1(0.1 + 1.10) = 1.10 + 0.1(1.20) = 1.10 + 0.12 = 1.22$.

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